

# On Development of the Negative Phrase in Ossetic\*

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The Modern Ossetic languages exhibit a typologically uncommon pattern of negative concord – a combination of obligatory negative spread and obligatory movement of Neg-phrases into an immediately preverbal position. I argue that this system developed historically from a fairly common phenomenon, the preverbal placement of focus. Contacts with South Caucasian languages, Georgian and Svan, must have played a crucial role in the emergence of this system. I propose a specific development scenario, according to which the negative movement is diachronically related to focusing, whereas the negative spread is an outcome of the fact that the negation markers got reanalyzed into the negative prefixes on wh-items.

## 1. Introduction

Since the work of Jespersen (1917), and the introduction of ‘Jespersen’s cycle’, the diachrony of negation has been attracting considerable attention from theoretical linguists, (Van Gelderen 2008, Larrivé 2011, Willis et al. 2013, Mosegaard Hansen and Visconti 2014, Zeijlstra 2016, and references there). However, the emergence of the negative phrase ‘from the scratch’ has been described so far in very few languages except English (Kemenade 1997, Haeberli and Ingham 2007, Haeberli 2011).

In this paper, I explore such a development in Ossetic. Ossetic is a cover term for two closely related Eastern Iranian languages, Iron and Digor, spoken in the Central Caucasus. Although there are subtle yet interesting differences between the negation system of Digor and Iron, for the purposes of this paper the two languages behave identically. The Ossetic languages exhibit a typologically uncommon pattern of negative concord – a combination of obligatory negative spread (which by itself is cross-linguistically rare) and obligatory movement of Neg-phrases into an immediately preverbal position.

I argue that this system developed historically from a fairly common phenomenon, the preverbal placement of focus. Under the proposed scenario, the negative movement is diachronically related to focusing, whereas the negative spread is an outcome of the fact that the negation markers got reanalyzed into the negative prefixes on wh-items. I will argue that the NegP in Ossetic emerged as a result of splitting of the FocP into the focus phrase in the proper sense and a lower NegP.

In Ossetic, negative indefinites are formed by prefixing a negative marker to a wh-word, e.g., the Digor Ossetic *ne-ʃi* NEG-what ‘nothing’, *ne-kəd* NEG-when ‘never’, etc. In a clause, all the negative indefinites or the sentential negation marker have to be placed in the immediately preverbal position (1 a-c). The sentential negation markers are incompatible with negative indefinites (1d). Unless explicitly indicated otherwise, all Ossetic examples in this paper are from Digor.

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\*The Ossetic data for this paper have been collected in the course of my field work in North Ossetia in 2007 – 2021. I thank everybody who helped me to conduct my research on Ossetic for their patience and generosity. I thank Nicholas Sims-Williams and Leonid Kulikov for their help in analyzing Middle Iranian and Vedic data. I thank the audiences at DIGS13, Generative Linguistics in Poland 7, and Lectures on Iranian Linguistics in Arizona for very useful feedback to earlier versions of this paper, and the anonymous reviewers for their comments. Collection of the Svan data was supported by ISF grant 2276/23.

- (1) a. mēdinə soslan-i **nə** warzuj  
M. S-ACC<sup>2</sup> NEG loves  
'Madina does not love Soslan.'
- b. soslan-i **ne-kəd** **ne-ke** warzta  
S-ACC NEG-when NEG-who loved  
'Nobody ever loved Soslan.'
- c. \***ne-kəd** **ne-ke** soslan-i warzta  
NEG-when NEG-who S-ACC loved  
*Idem* (intended reading)
- d. \*soslan-i **ne-kəd** **ne-ke** **nə** warzta  
S-ACC NEG-when NEG-who NEG loved  
*Idem* (intended reading)

It is natural to inquire as to how such a system came into existence. Erschler (2010) argued that it is contacts with Georgian and Svan, neighboring South Caucasian languages, that probably influenced the Ossetic negation system.

I propose a specific diachronic scenario for the emergence of this pattern of negation, starting with the situation in the attested Old Iranian languages, as described by Briceño Villalobos (2019). I hypothesize that the preverbal placement of negation started out as adjunction to a preverbal FocP. Later on, the position of negation was reanalyzed into a dedicated NegP. This development must have occurred across all Iranian languages, including an ancestor of Ossetic. However, due to South Caucasian influences, in Ossetic, the syntax of indefinites in the scope of negation has become strikingly different from that in the rest of Iranian.

More specifically, I argue for the following scenario. At the initial stage, common for all Old Indo-Iranian languages, the sentential negation marker was an adverb carrying an interpretable feature [iNeg]. It could be adjoined to the clause-initial FocP. Only one negative indefinite, 'no one,' existed. In the capacity of indefinites in the scope of negation, wh-based polarity-neutral indefinites were used. They could be fronted into the clause-initial focus position, thus becoming adjacent to the negative adverb (2).

- (2) Neg<sup>[iNeg]</sup>=[<sub>FocP</sub> Id<sub>f1</sub> Id<sub>f2</sub> [...VP]...]

At the second stage, common to all Middle Iranian languages, the fronting of negation was gradually lost, and simultaneously there emerged a preverbal focus projection. After this stage the ancestor of Ossetic parted ways with the rest of Iranian languages so far as the development of negation is concerned.

At the third, Ossetic-specific stage, the negative adverb got adjoined to the now preverbal FocP, whereas the indefinites in the scope of negation were obligatorily moved into the specifiers of the focus projection. The negative adverb still carried an interpretable Neg feature, (3).

- (3) [fronted material ... Neg<sup>[iNeg]</sup>=[<sub>FocP</sub> wh<sub>1</sub> wh<sub>2</sub> [V ...]...]

At the fourth stage, the preverbal FocP split into a higher FocP and NegP below it. Wh-items in their capacity of NPIs got endowed with an uninterpretable [uNeg] feature, (4), which drove their movement into NegP.

- (4) [fronted material ... [<sub>FocP</sub> X [<sub>NegP</sub> Neg<sup>[iNeg]</sup>=wh<sub>1</sub><sup>[uNeg]</sup> wh<sub>2</sub><sup>[uNeg]</sup> [V...]]...]

<sup>2</sup> Glosses and abbreviations: ABL ablative; ACC accusative; ALL allative; CNV converb; CON conjunctive; CTR contrastive topic marker; DAT dative; EMP emphatic; FUT future; HAB habitual; IDF indefinite; IMP imperative; INF infinitive; LOC locative; NC Negative Concord; NEG negative; NI Negative indefinite; NOM nominative; NPI negative polarity item; OBL oblique; PL plural; POSS possessive; PRF perfective; PRS present; PRT participle; PRV preverb; PST past; SG singular; SUB subjunctive; SUP superessive. Old and Middle Iranian examples are reproduced as they appear in the original publications with glosses added by me.

Then the negative item procliticized to the chain of *wh*-items got reanalyzed into a prefix. That triggered the final, fifth, stage: the [uNeg] feature on the indefinites received an overt spell-out, thus giving rise to the modern system. A negative operator high in the structure became the carrier of the [iNeg] feature.

- (5) [Op<sub>Neg</sub> [fronted material....[<sub>FocP</sub> X [<sub>NegP</sub> NEG-*wh*<sub>1</sub><sup>[uNeg]</sup> NEG-*wh*<sub>2</sub><sup>[uNeg]</sup> [V...]]...]

An obvious disclaimer is that in the absence of direct evidence any such analysis is unavoidably highly speculative.

The paper is organized as follows. In Section 2, I recall the notions and terms related to negation and negative concord, and sketch the synchronic analysis of negative concord to be used in the paper. In Section 3, I review the relevant basic facts from Ossetic grammar and the history of the language. In Section 4, I introduce the relevant facts about negation in Modern Ossetic and lay out the analysis adopted here. In Section 5, I demonstrate the parallels between Ossetic and South Caucasian languages, and argue why using South Caucasian evidence is legitimate in a diachronic analysis of Ossetic negation. In Section 6, I address negation in Modern Iranian languages other than Ossetic. In Section 7, I present the proposed scenario. In Section 8, I provide evidence from Georgian and Svan to argue for the contact-induced stages of the proposed scenario. In section 9, I discuss and reject a possible alternative scenario.

## 2. Generalities on negation and Negative Concord

In this section, I recall general negation-related facts, definitions, and terminology that will be relevant for the rest of the paper, and sketch the analysis of Negative Concord I adopt.

### 2.1 Concepts and definitions

I will call the **sentential negation** the morphological expression of negation in a clause, following the usage of Miestamo (2005). If the sentential negation has a dedicated position in the syntactic structure of the clause, this position will be called **NegP**, as it has become standard since Pollock (1989).

I adopt the following standard assumption about the syntax-semantics interface (6).

- (6) For an operator *Op* to take scope over a constituent *XP*, *Op* must c-command *XP*.

The operator relevant for us in this paper is negation. As we will see, the location of the semantic negation operator in the structure does not have to coincide with the locus of sentential negation.

A key issue for this paper is the morphosyntactic behavior and semantics of **indefinites in the scope of (semantic) negation** and their interaction with the sentential negation. While not attempting to present a comprehensive typology, for which the reader is referred, e.g., to (Van der Auwera and L. Van Alsenoy 2016, 2018, Van der Auwera and Krasnoukhova 2020), I will focus here on phenomena directly relevant to this paper. Before addressing the relevant possible strategies, it is worth noting that several such strategies may co-exist in a single language. Informally speaking, the question I am asking is about how indefinites and the sentential negation will behave in the translational equivalent of (7) in a given language.

- (7) No one recognized anyone.

Possible strategies include first, using semantically negative indefinites, i.e. ones that have the negative force of their own (the English *no one*, *nothing*, etc). In this case, they cannot co-occur with each other lest their negative forces cancel out, as it happens e.g. in (8).

- (8) No one recognized no one. = Everyone recognized someone.

Second, intrinsically non-negative indefinites can be used in the scope of negation. Those can be Negative Polarity Items (NPIs), i.e. require a downward entailing environment to be licensed, as it is the case for the English *anyone*, *anything*, etc. (Tovena 2020, a.o.). However, in some negation systems, indefinites with a wide distribution can be used for these purposes. Not uncommonly these are wh-phrases, see, e.g., Tran and Bruening (2013: 219) for Vietnamese (9); Erschler (2023) for indefinites used with the colloquial emphatic negation in Russian; and a general discussion of wh-words used as wide-purpose indefinites in Hengeveld et al. (2022).

(9) Vietnamese, Tran and Bruening (2013)

Tân không gặp ai.  
 Tan NEG meet who  
 ‘Tan does/did not meet anyone.’

Finally, negative indefinites may exhibit **Negative Concord** (NC), see Giannakidou (2020) and references there for a recent overview. In this case, negative indefinites ostensibly have negative force of their own, because they can serve as negative fragment answers (10a); but they co-occur with each other without creating a double negation effect (10b).

(10) Russian

a. Q: kto ponimaet jevo stat'ji? A: ni-kto  
       who understands his articles NEG-who  
       ‘Who understands his articles?’ ‘No one.’

b. jevo stat'ji ni-kto ni-kogda ne=ponimaet  
       his articles NEG-who NEG-when NEG-understands  
       ‘No one ever understands his articles.’

I will call the indefinites participating in NC **Neg-phrases**. NC languages vary in how neg-phrases interact with the sentential negation. Specifically, a language may exhibit **Strict Negative Concord** (the term proposed by Giannakidou 1997), i.e. Neg-phrases will require the presence of sentential negation (10b), or **Negative spread** (the term of Den Besten 1986). In the latter case, Neg-words may co-occur with each other, but the sentential negation must be absent (11).

(11) Drechterland Dutch, Den Besten (1986: 205)

Ik win nooit niks  
 I gain never nothing  
 ‘I never gain anything.’

NC languages present an obvious analytic puzzle: Neg-phrases seem to both have and have not negative force of their own. Furthermore, languages that use intrinsitically non-negative indefinite, may have such an indefinite in the scope of negation but obviously not c-commanded by the sentential negation (10b). If one expects the sentential negation to bear negative force, or in other words be the spellout of the negative operator, this fact contradicts the principle in (6).

## 2.2 Negative Concord: Idea of Analysis

A number of proposals have been advanced to analyze NC, see an overview in Giannakidou (2020) and the references there. It is beyond the purposes of the present paper to lay out all these approaches, let alone to adjudicate between them. In this section, I will only sketch the approach I am going to adopt in my analysis.

An idea advanced independently by several authors (e.g., by Brown and Franks (1999) for Russian and by Karimi (2005) for Persian) is to separate the morphological expression of negation (hosted by the NegP) from a negative operator contributing the negative semantics (12). Zeijlstra (2004) worked out a general formalism, which uses the machinery of interpretable and uninterpretable negative features.

The informal meaning of this approach is that all negative markers are interpreted as one instance of negation, i.e., one negative operator. Given that none of the negative items seems to play a distinguished role, nor does their scope necessarily correspond to their surface position, it is a natural theoretical move to posit a phonologically null element with negative semantics, a negative operator.

On this type of analysis, the relation between the operator and the NegP is either Agree (as in the proposals of Brown and Franks, Karimi, or Zeijlstra) or some kind of semantic licensing (Preminger and Polinsky 2015).

(12)    [... Op(iNeg)                      [NegP(uNeg) ... [vP...]]...]

Neg-phrases, on this type of analysis, are semantically non-negative, but are licensed by a null negative operator. In Zeijlstra's (2004, 2008) analysis, they bear a [uNeg] feature. While the original proposal of Zeijlstra's creates an issue with the directionality of Agree (Preminger 2013, Preminger and Polinsky 2015, Rudnev 2021, Bárány and van der Wal 2022, Keine and Dash 2023). In this paper, nonetheless, I adopt Zeijlstra's approach<sup>3</sup>.

An additional benefit of positing a null negative operator high in the structure is that it allows one to explain scope paradoxes where an indefinite in the scope of negation is not c-commanded by the sentential negation, see (30) in Section 6 below for specific examples. In languages that exhibit such interaction between sentential negation and indefinites, the negative operator is high, while the sentential negation marker lacks negative force of its own.

To repeat, this is the theoretical framework I adopt for the diachronic considerations in this paper.

### 3. Background on Ossetic

Given that according to the scenario I propose in this paper, the contacts between Ossetic and South Caucasian languages played a crucial role in the development of Ossetic negation, the overall linguistic contact history of Ossetic is relevant for our discussion. In this section, I briefly overview the contact history, and introduce the necessary facts about the grammar of Ossetic.

#### 3.1 History

Ossetians are descendants of the medieval Alans and, more distantly, of Sarmatians (a people related to the Scythians), Alemány (2000: 8-9). For about a millennium or even longer, the ancestors of modern Ossetians were nomads of the Eurasian steppes, whose migration territory stretched between the Danube river in the West and the Altay mountains in the East, Windfuhr (2009: 8). For nearly two millennia, Ossetians have lived in the Caucasus and apparently have had no contact with other Iranian peoples, who have occupied a more or less contiguous area elsewhere. I will use the term Proto-Ossetic to refer to all earlier stages of these languages once their ancestor split from the rest of the Iranian.

Historical phonology and morphology of Ossetic have been actively studied (Cheung 2002, 2008, Kim 2003, 2007, 2022a, 2022b, 2023, and references there). There has been much less research on historical syntax of the language.

Due to the nomadic past, virtually any North Eurasian language could be a source of contact influence on (Proto-)Ossetic. For the last two millennia, it is the autochthonous languages of the Caucasus that played a particularly significant role. Probably the best known and most telling example of Caucasian influences is the fact that Ossetic acquired a series of ejectives: *p'*, *tʰ'*, *ʃ'*, *t'*, *k'*. The contacts left considerable loanword strata as well, Thordarson (2009: 22-64) and references there.

In Erschler (2009), it was proposed that the Ossetic possessive proclitics/prefixes developed from Iranian second position clitics under West Caucasian influences. Erschler (2010) argued that the morphology of negative indefinites in Ossetic was influenced by South Caucasian languages. Erschler

<sup>3</sup> The formalism developed in Deal (2022a, 2022b) allows one to sidestep the directionality issue and I assume that the analysis adopted here, as well as other analyses of this type, can be straightforwardly reformulated in Deal's framework.

(2012) proposed that the pattern of *wh*-movement in Ossetic was influenced by South Caucasian languages as well. The latter paper proposes a specific scenario for this development.

Any diachronic treatment of Ossetic faces a number of challenges. First, there exist virtually no written sources that would reflect earlier stages of Ossetic grammar (Thordarson 2009: 188, Lubotsky 2015). First connected texts in Ossetic date from the late 18th and early 19th c., and already show an essentially modern state of the language. Because of that, to gain any insight into the history of Ossetic, one has to carefully look at the related Iranian languages and various unrelated languages of Eurasia that Ossetic potentially could have contact with. In favorable cases, that, along with typological considerations, allows us to advance a plausible conjecture about the respective diachronic development.

### 3.2 An Overview of Ossetic Grammar

The Ossetic languages are predominantly head-final, with mostly agglutinative morphology. For recent grammar sketches see Erschler (2018, 2020). Ossetic possesses a moderately rich case system, three tenses (present, past, and future), and three moods: indicative, subjunctive, and imperative. Verbs may carry preverbs that express spatial and aspectual meanings. Both languages have a rich system of enclitic pronouns and adverbs. In Iron, most enclitics obligatorily occupy an (appropriately defined) second position in the clause, whereas in Digor this restriction is more lax. The SOV order is statistically predominant, but actually the constituent order is usually relatively free. Sentences tend to be verb-final.

Ossetic shows obligatory multiple *wh*-movement into a preverbal position (13a), (Erschler 2012, Borise and Erschler 2022). *Wh*-phrases must be adjacent to the predicate, as illustrated by the contrast between (13b) and (13c), and can be only separated by certain items, including negation markers and negative phrases<sup>4</sup> (13d).

- (13) a. *ɛʒinɐ*                      *soslan-en*                      **ka**                      **ʃi**                      *ravardta?*  
yesterday                      Soslan-DAT                      who                      what                      give.PST.3SG  
‘Who did give what to Soslan yesterday?’
- b. *soslan-i*                      **ka**                      *warzuj?*  
soslan-ACC                      who                      love.PRS.3SG  
‘Who loves Soslan?’
- c. \***ka**                      *soslan-i*                      *warzuj?*  
who                      soslan-ACC                      love.PRS.3SG  
*Idem* (intended)
- d. **ka**                      [*neʃi kiwunugɐ*]                      *ba-kastɐj?*  
who                      nothing book                      PRV-read.PST.3SG  
‘Who did not read any book?’ (lit. ‘Who read no book?’)

Topics are placed clause-initially, as illustrated in (14a) for *ɛχtsa* ‘money’, and in (14b) for the contrastive topic *mɛnɛn* ‘me’.

- (14) a. **ɛχtsa**=*nɛmɐ*                      *k’apek=der*                      *ne-kutɐj*                      *tsudɐj*  
money=ALL.1PL                      kopeck=EMP                      NEG-where.ABL go.PST.3SG  
‘As for money, we didn’t get a red penny from anywhere. Aghuzarti 2008: 20
- b. *kader*                      *niwaz-gɐ*                      *kodta*                      [**mɛn-en**=*ba mɛ=raɕ*  
someone                      drink-CNV                      do.PST.3SG                      I.OBL-DAT=CTRPOSS.1SG=back  
*is-χawidɐ*                      *don-i*                      *χɛssun-ɐj*  
PRV-fall.PST.HAB.3SG                      water-GEN                      carry-INF-ABL  
‘Somebody drank, and I had my back chafed from carrying water.’ (from a recorded text)

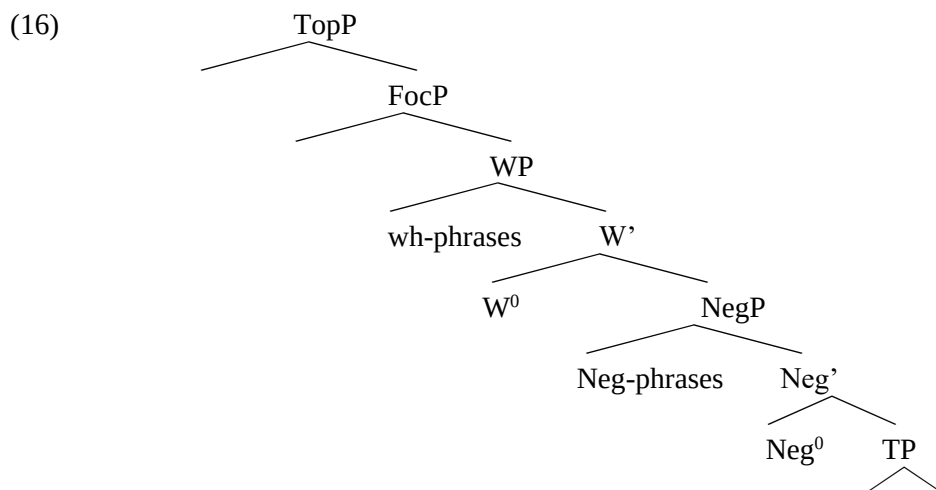
<sup>4</sup> If a negative indefinite is embedded in a DP, as in (13d), the entire DP is placed in the preverbal position. I call such constituents negative phrases.

Foci are either preverbal or postverbal. The existence of these positions can be inferred from the word order in sentences with items associated with focus (see a detailed list of these, e.g., in Beaver and Clark 2008: 52-53, 82) and native speakers' judgements on word order in question-answer pairs. Sentences in (15) provide evidence for preverbal focus. A particular clear source of evidence comes from constituents associated with 'only'. It is uncontroversial that such constituents must be associated with focus (Beaver and Clark 2008), and, in Ossetic, *ermest(der)* 'only' must immediately precede the constituent it is associated with. The preverbal placement of focused items is illustrated by the contrast between the grammatical sentence in (15b), where the constituent associated with 'only' is immediately preverbal and the ungrammatical sentence in (15c), where it is separated from the verb by another DP.

- (15) a. Free choice indefinites  
 woj wəlemχasən=ba=ma=də ɛχsəvə  
 that in.addition=CTR=else=ACC.2SG night  
 [kəʃʃifendi afoni]<sub>F</sub> [ɛr-iχal kənun kodtajdɛ] direktor  
 whatever time PRV-awakeness do.INF do.HAB.3SG principal  
 'On top of that, the director used to wake you up ANY TIME at night.'
- b. constituents associated with 'only'  
 [ermest medine]<sub>F</sub> ravardta kiwunugə soslan-ən  
 only Madina give.PST.3SG book Soslan-DAT  
 'It is only Madina who gave a book to Soslan.'
- c. \*[ermest medine]<sub>F</sub> kiwunugə ravardta soslan-ən  
 only Madina book give.PST.3SG Soslan-DAT  
*Idem* (intended reading)
- d. ɛr-izadəj=ma=ʃi [ermestder dɛs]<sub>F</sub>  
 PRV-remain.PST.3SG=still=ABL.3PL only ten  
 'Of them, there still remained only ten.' Maliti (2006)

The existence of a postverbal focus will serve as one of the grammatical parallels between Ossetic and Georgian. While its structural analysis is orthogonal to our current purposes, Borise (2023) advances a specific proposal about the Georgian postverbal focus.

I will adopt here the structure of the Ossetic clause proposed in Borise and Erschler (2022) (16).



#### 4. Negation in Ossetic: The Synchrony

In this section, I lay out basic facts about negation in Ossetic, and present a synchronic analysis based on the idea sketched in Section 2.2. These will be compared in Section 6 to the system in the rest of Modern Iranian.

#### 4.1 Basic Data

Sentential negation<sup>5</sup> in Ossetic is expressed either by a mood-dependent proclitic on the verb (17 a, c), or by negative indefinites (17 b, d). Both sentential negation markers have solid Iranian etymologies, there was no renewal similar to what occurred in many Germanic languages, French, or Greek (Van Gelderen 2008).

Morphologically, negative indefinites are composed of a negative prefix and a *wh*-stem. As illustrated in (17), the negative prefix on negative indefinites depends on the mood exactly in the same manner as the sentential negation marker. In (17a) and (17b), the verbs are in the indicative, and accordingly the negation marker in (17a) is *nɐ* and the negative prefix on the negative indefinite in (17b) is *ne-*. On the other hand, the verbs in (17c) and (17d) are in the imperative, and the sentential negation marker and the negative prefix have the form *ma*.

- (17) a. Negative marker in an indicative clause  
tikis mist-ɐj **nɐ**=tɛrs-uj  
cat mouse-ABL NEG=fear-IND.PRS.3SG  
‘The cat doesn’t fear the mouse.’
- b. Negative indefinite in an indicative clause  
i tikis **ne**-kɛmɐj tɛrs-uj  
this cat NEG-who.ABL fear-IND.PRS.3SG  
‘This cat fears of no one.’
- c. Negative marker in an imperative clause  
**ma**=tɛrs-ɐ  
NEG.IMP=fear-IMP.2SG  
‘Don’t fear!’
- d. Negative indefinite in an imperative clause  
**ma**-kɛmɐj **ma**-kɛd tɛrs-ɐ  
NEG.IMP-who.ABL NEG.IMP-when fear-IMP.2SG  
‘Don’t be ever afraid of anyone!’

Negative indefinites are compatible with each other (18a) but not with the sentential negation marker, (18b), that is to say, Ossetic shows obligatory negative spread.

- (18) a. **ne**-kɛmɐ **ne**-tʃi iʃ-ʃerdton  
NEG-who.ALL NEG-what PRV-find.PST.1SG  
‘I didn’t find anything at anybody’s place.’ Aghuzarti 2008: 62
- b. soslan-i **ne**-ke (\***nɐ**) warzuj  
S-ACC NEG-who NEG love.PRS.3SG  
‘Nobody loves Soslan.’

Apparently, obligatory negative spread is rather uncommon cross-linguistically, (Giannakidou 2000:460, Van der Auwera and Van Alsenoy 2016). Another remarkable feature of the Ossetic negation system is the obligatory negative movement<sup>6</sup>. All negative indefinites in the clause have to be placed in the immediately preverbal position, as illustrated by the contrast between the

<sup>5</sup> A more detailed description of negation in Digor can be found in Erschler and Volk (2011).

<sup>6</sup> A reviewer suggests that *wh*-phrases and *neg*-phrases occupy the same position in Ossetic, and the presence of *neg*-movement is parasitic upon *wh*-movement. In actuality, these are two different phenomena. *Neg*-movement and *wh*-movement have different landing sites. A language that exhibits obligatory *wh*-movement and has *neg*-phrases built upon *wh*-phrases, may nonetheless lack *neg*-movement. Such, for instance, are Slavic languages. Furthermore, neither *neg*-movement, nor *wh*-movement in Ossetic are synchronically instances of focus movement. Thus, considerations from Turkish of Göksel and Özsoy (2000) are not applicable in the case of Ossetic.



grammatical sentence in (19a), where this requirement is fulfilled, and the ungrammatical ones in (19b-d), where it is not.

- (19) a.    soslan-i **ne-kəɖ**            **ne-ke**            warzta  
           S-ACC NEG-when        NEG-who        love.PST.3SG  
           ‘Nobody ever loved Soslan.’  
       b.    \***ne-kəɖ**            **ne-ke**            soslan-i            warzta  
           NEG-when        NEG-who        S-ACC            love.PST.3SG  
           *Idem* (intended reading)  
       c.    \***ne-ke**            soslan-i warzta            **ne-kəɖ**  
           NEG-who        S-ACC love.PST.3SG        NEG-when  
           *Idem* (intended reading)  
       d.    \*soslan-i            **ne-ke**            warzta            **ne-kəɖ**  
           S-ACC            NEG-who        love.PST.3SG        NEG-when  
           *Idem* (intended reading)

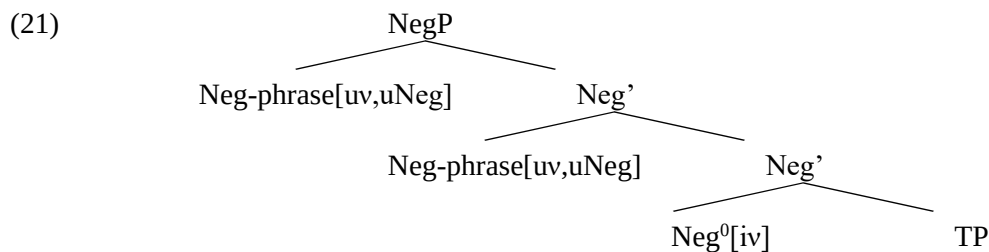
The only items that may intervene between the negation marker or negative indefinites, and the verb are Wackernagel enclitics, if the complex of negative items and a verb happens to be the first in the clause<sup>7</sup>. This is illustrated for the 3sg accusative enclitic *=(v)j* in (20).

- (20) a.    *ne=j*                            *ɐvzaruj*  
           NEG-who=ACC.3SG        choose.PRS.3SG  
           ‘He cannot figure it out.’ Aghuzarti (2008: 76)  
       b.    *ne-ke=j*                    *ba-jjafɖɛnɐj*  
           NEG-who=ACC.3SG        PRV-reach.FUT.3SG  
           ‘No one will catch up with it.’ Xoxojti (2011: 143)

It is natural to inquire about how this obviously unusual pattern of negation and negative concord developed historically.

## 4.2 Synchronic Analysis

I will adopt here the analysis of the Ossetic negation from (Erschler and Volk 2011; Borise and Erschler 2022). It implements the approach sketched in Section 2.2. According to this analysis, NegP is taken to exist and to immediately dominate the TP (21).



The negation marker and negative indefinites are XPs carrying a strong uninterpretable NEG-feature [uNeg]. Besides that, I posit that Neg<sup>0</sup> bears a formal feature [iv], while NIs bear the matching feature [uv]. The need to check it drives the movement of negative indefinites into NegP. The sentential negation marker *ne/ma* is spelled out as the last resort to satisfy the EPP feature of Neg<sup>0</sup> in

<sup>7</sup> If several negative indefinites appear clause-initially, enclitics are normally placed after the first NI.

(i)    *ne-ke=in*            *ne-ʃɪ*            *zaxta*  
       NEG-who=DAT.3SG        NEG-what        say.PST.3SG  
       ‘No one said anything to her/him.’

the absence of neg-phrases<sup>8</sup>. Here, I take EPP to be an essentially phonological requirement (Holmberg 2000, Ndayiragije 2000, Miyagawa 2001, 2005, Bobaljik 2002, Takahashi 2002, and Holmberg and Hróarsdóttir 2003). The adjacency between Neg-phrases and the verb, as was illustrated in (12), is achieved by the verb head-moving into Neg<sup>0</sup>, whenever NegP is merged<sup>9</sup>. I assume that any material preceding NegP is moved there for information structure reasons, i.e. focusing or topicalization.

The negative operator, on the other hand, must be situated very high in the clause, given that Ossetic exhibits negative conjunctions (*ne-dər* in Digor, and a richer inventory in Iron, Erschler 2013) that do not create double negation effects with Neg-phrases or sentential negation markers (22a). On the other hand, they do not need the presence of sentential negation marking or Neg-phrases to create negative force (22b).

- (22) a. **nedər**=fɪ                      neɟɪ    ra-gəldzənə  
neither=ABL.3PL                nothing PRV-throw.FUT.2SG  
**nedər**=ba=səmə                neɟɪ-bal                ba-ftawudzənə  
neither=CTR=ALL.3PL        nothing-more        PRV-add. FUT.2SG  
'Neither will you subtract anything from them, nor will you add anything to them any more.' Xoxojti (2011: 113)
- b. **nedər**    turkə-mə                ra-χezdzənə                **nedər**    medəmə  
neither    yard-ALL                PV-move.FUT.2SG                neither    inside  
ba-tsəwudzənə  
PV-go.FUT.2SG  
'You will neither go out to the yard, nor enter inside.' Ik'ati (2011: 49)

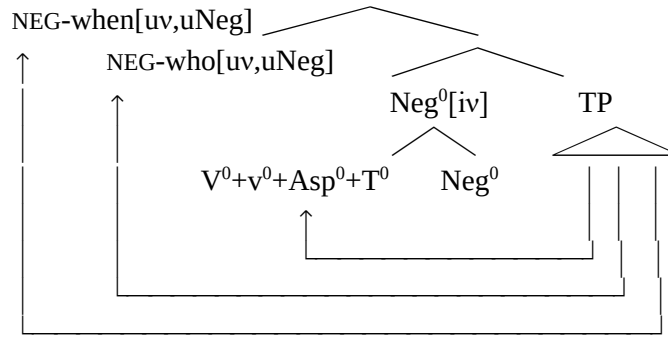
I illustrate the system with the derivation of (19a) repeated here as (23a). For the sake of simplicity, I take the negative operator to be an X<sup>0</sup>. I do not show the topicalization of the direct object *Soslan* to reduce clutter. The neg-phrases NEG-when[uNeg] and NEG-who[uNeg] move into the multiple specifiers of NegP and undergo Agree with Neg<sup>0</sup> in the feature [v]. The prefix *ne-* is the morphological manifestation of this agreement. Once Op[iNeg] is merged, it agrees with the items in NegP in the feature [Neg].

- (23) a.        soslan-i **ne-kəd**                **ne-ke**                warzta  
              S-ACC    NEG-when                NEG-who                loved  
              'Nobody ever loved Soslan.'
- b.                OpP  
                  /                \  
              Op[iNeg]                TopP  
                                      /                \  
                              Soslan-ACC                Top<sup>0</sup>                NegP

<sup>8</sup> I leave aside the issue of the dependence of negative markers on the modality. All what matters now is that all Ossetic negative markers behave identically in this respect.

<sup>9</sup> Accordingly, postverbal placement of direct objects (i a) and low adverbials (i b) is possible in negative sentences. However, given the fairly free word order in Ossetic, this fact by itself is not a strong evidence in favor of verb movement.

- (i) a.        eχsa-mə                gəsge=dər=imə                ne-ke                iɟ-ista  
              money-ALL                for.the.sake=EMP=ALL.3SG                NEG-who                PRV-give.PST.3SG  
              **e=k'ox**  
              POSS.3SG=hand  
              'Even for the sake of money, no one stretched their hand to him.' Gurdzibety (2006: 84)
- b.        soslan    ne-kəd                fə-dzəzoruɟ                **saburgaj**  
              Soslan    NEG-when                PRV-speak.PRS.3SG                slowly  
              'Soslan never speaks slowly.'



To recapitulate, Ossetic exhibits obligatory Negative Spread; the morphological marking of neg-words is modality dependent in the same manner as the sentential negation; and Neg-phrases are obligatorily moved into the specifiers of NegP, which is linearly immediately preverbal, and structurally immediately dominates the TP.

## 5. Parallels between Ossetic and South Caucasian

In this section, I make the case for extensive linguistic contacts between South Caucasian languages and (Proto-)Ossetic, and describe surface similarities between Ossetic, Georgian, and Svan morphosyntax of negation and negative indefinites, which will justify using Georgian diachronic data to argue for a specific diachronic scenario in Ossetic. Among the South Caucasian, it is Old and Modern Georgian and Svan that exhibit the most parallels with Ossetic and, for geographical reasons, are the most likely sources of contact influences.

Contacts between Alans and South Caucasian peoples are well attested historically, and, moreover, extensive lexical borrowing occurred in both directions (Abaev 1995: 373-393, Dzigicjty 2008: 210-211). Furthermore, non-trivial grammatical parallels exist between Ossetic, Georgian, and Svan. This allows us to conclude that significant linguistic contact between (Proto-)Ossetic and SC languages took place in the past. The direction of influences between (Proto-)Ossetic and SC languages is not as important for my current purposes as the conclusion that, as a result of these influences, some diachronic processes in (Proto-)Ossetic and Georgian must have proceeded in a similar manner.

Non-trivial grammatical similarities include a system of verbal prefixes with spatial and aspectual meanings (Thordarson 1982, Tomelleri 2009); the obligatorily preverbal placement of *wh*-phrases<sup>10</sup> (Harris 1984, Erschler 2012, 2015, and Borise 2023 for Georgian and Svan, Erschler 2012, Borise and Erschler 2023 for Ossetic); the existence of a post-verbal focus (Skopeteas and Fanselow 2010, Borise 2023 for Georgian, and Section 3.2 for Ossetic); the morphological structure of Neg-words, and, in the case of Ossetic and Svan, a pattern of preverbal placement of Neg-phrases<sup>11</sup>.

Svan and Georgian (in all attested historical stages of the latter) exhibit Neg-words. The morphology of these Neg-words is strikingly similar to what obtains in Ossetic, as was described in Section 4.1. They consist of a mood-dependent negative prefix and a typically *wh*-based stem, as illustrated in Table 1. Mood-dependent negative prefixes are nearly identical to the respective sentential negation markers, consider the pairs (24a-b), (24c-d) and (24e-f).

<sup>10</sup> In this respect, the other South Caucasian languages, i.e. Megrelian and Laz, behave identically to Georgian, Erschler (2012).

<sup>11</sup> Their placement in Georgian is rather different, see Borise (2023) for a detailed discussion.

- (24) Georgian
- a. ia        **ar**        muɟaobs  
       Ia        NEG        works  
       ‘Ia doesn’t work.’
- b.        **ara-vin**        (**ar**)        muɟaobs  
       NEG-who        NEG        works  
       ‘No one works.’
- c.        ia        **ver**        muɟaobs  
       Ia        NEG.POT        works  
       ‘Ia cannot work.’
- d.        **vera-vin**        (**ver**)        muɟaobs  
       NEG-who        NEG.POT        works  
       ‘No one can work.’
- e.        **nu**        muɟaob  
       NEG.IMP        work.2SG  
       ‘Don’t work!’
- f.        **nur-sad**        (**nu**)        muɟaob  
       NEG.IMP-where NEG.IMP        work.2SG  
       ‘Don’t work anywhere!’

Table 1. Negative indefinites in Ossetic, Georgian, and Svan

|                 | ‘nobody’                                       | ‘nothing’                                         | ‘never’                                           |
|-----------------|------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| Digor Ossetic   | <i>ne-/ma-ke</i><br>NEG <b>-who</b>            | <i>ne-/ma-ɟi</i><br>NEG <b>-what</b>              | <i>ne-/ma-kəd</i><br>NEG-/N.IMP- <b>when</b>      |
| Old Georgian    | <i>ara-/nu-/vera-vin</i><br>NEG- <b>-who</b>   | <i>ara-/nu-/vera-raj</i><br>NEG <b>-what</b>      | <i>ara-/nura-sada</i><br>NEG <b>-where</b> (sic!) |
| Modern Georgian | <i>ara-/nura-/ vera-vin</i><br>NEG- <b>who</b> | <i>ara-/nura-/ vera-peri</i><br>NEG <b>-color</b> | <i>ara-/nura-/vera-sodes</i><br>NEG- <b>ever</b>  |
| Upper Bal Svan  | <i>d-æ:r/n-/deɟ-(j)æ:r</i><br>NEG- <b>who</b>  | <i>de:sa/no:sa-/deɟ-ma</i><br>NEG- <b>what</b>    | <i>de:-/no:-šoma</i><br>NEG- <b>when</b>          |

Such a morphological structure of negative indefinites is relatively uncommon typologically, (Erschler 2010). In the Caucasus, it is only present in Ossetic, Georgian, Svan, and Tsova-Tush, a Northeast Caucasian language spoken in one village in Georgia, and heavily influenced by Georgian, Holisky and Gagua (1994, 2.4.7).

Modern Georgian allows Neg-phrases to co-occur with sentential negation (24 b,d,f), and does not require them to be immediately preverbal, Borise (2023). In Svan, Neg-phrases form either a preverbal chain, similarly to Ossetic, or a postverbal one, in which case a negative item must occur immediately preverbally, Erschler (2022). If the chain is preverbal, Neg-phrases may not co-occur with sentential negation (25).

- (25) Upper Bal Svan, Erschler (2022)
- gela-d **de:-mtɟik**        **de:-me**        **d-æ:r**        (\***de:m/\*ma:m**)        anundrevi:ne  
       G-ERG NEG-ever        NEG-where        NEG-who        NEG        offended  
       ‘Gela has never offended anyone anywhere.’

Moreover, unlike in Georgian (Borise and Polinsky 2018, Borise 2023), but similarly to Ossetic, the preverbal chain of Neg-phrases in Svan obligatorily intervenes even between a wh-phrase and the verb (16).

- (26) Upper Bal Svan
- wh NI ... NI V? / \*NI ... NI wh V?
- a.        xed        sopels        **dæ:r**        **de:mgwaf**        itre?

- which village-DAT nobody nothing drinks  
 ‘In which village does no one drink anything?’
- b. \*dæ:r de:mgwaf xed sopel-s (de:m) itre?  
 nobody nothing which village-DAT NEG drinks  
*Idem* (intended)
- c. mest’ias jær de:mʃik arda?  
 M-DAT who never has.been  
 ‘Who has never been to Mest’ia?’
- d. \*mest’ia-s de:mʃik jæ:r ma:m/de:sa arda?  
 M-DAT never who NEG has.been  
*Idem* (intended)

To recapitulate, the numerous non-trivial parallels between the synchronic grammars of Ossetic, Georgian, and Svan discussed in this section strongly suggest that significant language contacts took place in the past between these languages. Later, in Section 8, we will compare the Old Georgian and Svan systems with the Modern Georgian one to argue for specific diachronic developments in Ossetic. The premise is that some kind of co-evolution of contacting languages occurred.

## 6. Negation in Modern Iranian

In this section, I overview the negation systems across Modern Iranian languages to show that the Ossetic system is unique among them. This still leaves us with a possibility that the Ossetic system was inherited from Proto-Iranian, while the systems in the rest of the Iranian languages are all innovated. However, once we examine the Old Indo-Iranian negation in Section 7.1, we will be able to conclude that the Ossetic system must be an indepent innovation.

Modern Iranian languages all exhibit preverbal sentential negation; in some languages the negation markers are mood-sensitive. In complex verbs, negation is hosted by the light verb, as is illustrated for Persian in (27).

- (27) Persian, Taleghani (2006: 154)  
 zamin **na**-xord-am  
 ground NEG-eat-1SG  
 ‘I did not fall down.’

As indefinites in the scope of negation, intrinsically non-negative indefinites (28) or morphologically marked Neg-words (29) are used. In the latter case, Neg-words are usually marked with *hič-* or its cognates, possibly a Persian loan. Both systems can coexist in a single language. Unlike in Ossetic, no particular restrictions exist on the linear position of NPIs or Neg-words.

- (28) Wakhi, Erschler (2010:95)
- a. An intrinsically non-negative indefinite  
**i-ʃiz** tsə=kʃəji maz-i aga tsar  
 IDF-what if=hear:2SG I:OBL-ACC awakening do:2SG  
 ‘If you hear anything, wake me up.’
- b. The same pronoun in a negative clause  
**me** pʃad **i-ʃiz**  
 NEG.IMP touch:2SG IDF-what  
 ‘Do not touch anything!’
- c. Munji, Gruenberg (1972: 419)<sup>12</sup>  
**kdi** či-kərkint da qura ləghay-o  
 who NEG-be.able PREP hill climb-INF  
 ‘No one can climb this hill.’
- d. Pashto, Robson and Tegey (1996: 72)

<sup>12</sup> The glosses are mine.

- tsə**                      me                      **nə**                      di                      ke.ʃi  
something                      I                      NEG                      have                      done  
‘I haven’t done anything.’
- e. Persian, Van der Auwera and Koohkan (2022: 5)  
**kas-i**                      zang                      **na-zad-Ø**  
HUM.SG-IDF                      ring                      NEG-hit.PST-3SG  
‘Nobody called.’
- (29) a. Persian, Taleghani (2006: 15)  
**hič-kas**                      Sârâ-ro **ne-mi-šenext**.  
NEG-who                      S.-ACC NEG-ASP-know.PST.3SG  
‘Nobody knew Sârâ.’
- b. Pashto, David (2014: 347)  
**hits**                      **waxt**                      pə                      šahi                      taxt                      kxé-**ne-nāst**  
none                      time                      on                      royal                      stage                      AOR\sit-NEG-sit.PST-PST.3SG.M  
‘(Shah Mahmud disavowed his royal birthright and) never assumed the throne.’
- c. Balochi, Korn and Jahani (2009: 688)  
go:                      **ičkass**                      gis-ay                      ta:                      abara                      **na-da:t-un**  
with                      nobody house-GEN                      in                      word                      NEG-give-PRS.1SG  
‘I did not talk with anyone in the house.’

Systematic generative studies of negation in Iranian only exist for Persian (Karimi 2005, Taleghani 2008, Kwak 2010, Kahnemouyipour 2017). However, given that the systems in other Modern Iranian languages look very similar, the analyses of Persian are applicable to them, at least at a first approximation.

A high position of the negative operator can be inferred from the fact that negation can scope over indefinites in the subject position (30).

- (30) a. Persian, Van der Auwera and Koohkan (2022: 5)  
**kas-i**                      zang                      **na-zad-Ø**  
HUM.SG-IDF                      ring                      NEG-hit.PST-3SG  
‘Nobody called.’
- b. Pashto, Grünberg<sup>13</sup> (1987: 103)  
no                      xabara ye                      pə                      qasam sara                      ham                      **cok**                      **na-arwi**  
then                      word his                      in                      oath with                      also                      who                      NEG-believe  
‘(If he would get known as a liar) then no one would believe his words,  
even if he would take an oath.’

The sentential negation marker must be above the vP, given that the light verb, which hosts negation (27), is the spellout of v<sup>0</sup> (Folli et al. 2005). The position of the sentential negation with respect to the AspP probably depends on the language<sup>14</sup>. In Persian, the negation marker precedes the aspectual prefix (31a), whereas in Pashto the order is reverse (31b).

- (31) a. Persian  
[Op(iNeg) [TP [NegP(uNeg) [AspP [vP]...]]
- b. Pashto (cf. Roberts 2000: 55)  
[Op(iNeg) [TP [AspP [NegP(uNeg) [vP]...]]

<sup>13</sup> The glosses are mine.

<sup>14</sup> Admittedly this difference may be a result of postsyntactic reordering, see e.g. Embic and Noyer 2001 for a general discussion of this phenomenon. This is not relevant for our main purpose, which is to compare between this type of a system and the one obtained in Ossetic.

This system is obviously different from the one that obtains in Ossetic. In the next sections, I will show that the Ossetic system must be an independent innovation, and will propose a specific scenario of its development.

## 7. The Diachronic Scenario

In this section, I describe the proposed scenario of development of the NegP stage by stage and provide evidence and arguments for my hypothesis.

### 7.1 Stage I: Old Indo-Iranian

To repeat, nothing is known about the Old Iranian ancestor of Modern Ossetic. As a proxy, I use data from attested Old Indo-Iranian languages, namely Old and Young Avestan, Old Persian, and Vedic. Using both Old Iranian and Old Indic data is justified because, as the findings of Briceño Vilalobos 2019 show, the systems in Avestan (Old Iranian) and Vedic (Old Indic) were fairly close. It stands to reason to assume that the properties of negation in the Old Iranian ancestor of Ossetic were also reasonably similar to those in the attested Old Indo-Iranian languages.

In early Indo-Iranian languages negation markers were sentence-level adverbials, which tended to appear close to the beginning of a sentence. Only one negative item per clause was allowed<sup>15</sup>, Briceño Villalobos (2019). “The negation occurs usually before the word which is under its scope, either at the beginning of a clause (...), or after, e.g., a subordinator (...).” Jügel (2017: 560). This is illustrated in (32) for the most archaic attested Iranian language, Old Avestan.

- (32) Old Avestan
- a. Yasna 45.1, Geldner (1896: 155)  
**nōiṭ** daibitīm duš.sastiš ahūm mərəšiiāt  
 NEG second.ACC evil.teacher existence.ACC destroy.OPT.PRS.3SG  
 ‘May not the teacher of evil destroy the second life.’
- b. Yasna 30.6, Geldner (1896: 107)  
 aiiā **nōiṭ** ərəš vīšiiātā daēvā-činā  
 this.DU.GEN NEG right discriminate Daevas-FOC  
 ‘The Daevas do not discriminate well between these two (spirits)’
- c. Yasna 43.12, Geldner (1896: 145)  
 aṭ tu=mōi **nōiṭ** asruštā pairiiaoyžā  
 and you=I.DAT/GEN NEG disobedience.LOC contradicted  
 ‘Thou hast never contradicted Me in disobedience.’ (transl. of Insler 1975)

The Old Indo-Iranian only had one semantically negative indefinite, wh-based ‘no one’, Briceño Vilalobos (2019: 129).

- (33) Old Avestan, Yasna 34.7, Geldner (1896: 125)  
**naēčīm** tām aniiēm yūšmaṭ vaēdā  
 nobody.ACC this.ACC other.ACC you.PL.ABL I.know  
 ‘In truth, I do not know at all anyone other than you /  
 In truth, I know no one other than you.’

Negative indefinites in the negation scope were (mostly) wh-based, (Briceño Vilalobos 2019: 132-142).

<sup>15</sup> Modulo the scarcity of the surviving data. For instance, “Old Avestan attests four instances of the negative indefinite *naēciš*.” (Briceño Vilalobos 2019: 159).

- (34) a. Old Avestan; Yasna 31: 18  
**mā**                      **čiš**      aṭ      vā      drəgvatō      (...)      gušta  
NEG.IMP                  who      then      EMP      deceitful                      listened  
‘No one at all who belongs to the deceitful listened<sup>16</sup>’ Geldner 1896: 113
- b. Young Avestan; Yasna 9: 21  
**mā**      **ciš**      paurvō                      būiḍyaēta              nō  
NEG      what      beforehand              notice                      us  
‘Let nobody notice us beforehand.’ Geldner 1896: 44

Following Kiparsky 1994 and drawing on the proposals of Hale 1987, I assume the following clause structure in archaic Indo-European languages (35). The items occupying Spec,TopP and Spec,FocP in (35) move there from their initial positions lower in the clause.

- (35) [TopP FocP [... V]...]

Given that some material could precede the fronted negative marker (32 b-c), it is reasonable to conclude that the negative marker was adjoined to FocP (36). Given that only one instance of negation was possible per clause, it is reasonable to assume that the negative marker (and the negative indefinite *naēciš* or its cognates) were semantically negative, i.e. bore [iNeg] feature.

- (36) Neg<sup>[iNeg]</sup>=<sub>[FocP</sub> Indefinite [...VP]...

Therefore, at this stage there is no need to posit any special features on negative polarity items that should be checked by moving close to the negation, their movement, when occurred, was just an instance of focusing.

## 7.2 Stage II: Later Old Iranian and Middle Iranian

At the next stage, the fronting of negation got gradually replaced by its preverbal placement. Concomitantly, a preverbal focus position emerged across Middle Iranian languages. I hypothesize that the preverbal placement of negation started out as adjunction to the now preverbal FocP. Later on, the position of negation was reanalyzed into a dedicated NegP. This development must have occurred across all Iranian languages, including an ancestor of Ossetic. However, due to South Caucasian influences, in Ossetic, the syntax of indefinites in the scope of negation has become strikingly different from that in the rest of Iranian.

The loss of negation fronting was already on the way in the less archaic attested Old Iranian languages, Young Avestan and Old Persian, where the negation marker starts appearing in the preverbal position (37). This was predominantly the case in Old Persian (Hale 1988: 34, Briceño Villalobos 2019: 128).

- (37) Young Avestan, Yasht 10.3, Gershevitch (1959: 74)  
yōi      miθrəm              **nōit**      aiwi-družinti  
REL      contract.ACC      NEG      break.PRS.3PL  
‘who do not break a contract’

Nonetheless, the placement of negation marker did not become rigidly preverbal until relatively late. The fronting of the negation marker was still possible in some Middle Iranian languages, as shown for Old Khotanese in (38a) and Sogdian<sup>17</sup> in (38b). The sources of these examples can be safely assumed to be originally written in the respective languages. *Book of Zambasta*, from which (38a) is taken, is believed to be originally written in Khotanese, (Emmerick 1979: 43). The Mugh

<sup>16</sup> The translation from Insler (1975: 41). Insler suggests that the negative marker *mā* should be replaced here by *naē*, but this is immaterial for our purposes.

<sup>17</sup> Sogdian was written in an Aramaic-based vowelless alphabet, with some Aramaic words used as ideograms for Sogdian ones. This is the case, for instance, with the negation *L*’.



letters, the source of (38b), are business letter coming from an 8th c. archive of a Sogdian governor, Livhits (2008: 12). Therefore, the clause-initial placement of the negators *ma* and *L'* in (38) is in all likelihood a feature of Old Khotanese and Sogdian grammars rather than a calque from some other languages.

- (38) a. Old Khotanese, Emmerick (1990: 395)  
**ma** buddha-dharma aysu upevīñi  
 NEG.IMP buddha-dharma I produce.SUB.1SG  
 'May I not produce the buddha-dharmas.' Book of Zambasta, 7.6
- b. Sogdian, Mugh letters, Livshits (2008: 62)  
 rt=kδ m'xy'n prymyδ yw'r rsth L' kwn't  
 and=if Maxyan according.to.it condition right NEG do  
 [L' m'yδ 'rtkw 'spwrnw δbr't]  
 NEG this flour.ACC completely give.CON.3SG  
 'And if Maxyan will not observe this condition, and will not give all the flour...'

Simultaneously, the replacement of an (essentially) clause-initial focus by a preverbal focus was underway in the Middle Iranian. The reasons for this change are unclear, but we have plenty of Iranian evidence for it. Wh-fronting, which was typical for Old Iranian, (Hale 1987), in attested Middle Iranian languages gets gradually replaced by another strategy, namely, putting them in the preverbal position (this is so in Sogdian, Middle Persian, and, to a lesser extent, in Khotanese). In (39), I illustrate this with Sogdian data, where clause-initial placement of wh-phrases is still attested (39c), but preverbal placement prevails (39a-b).

- (39) Preverbal wh
- a. Buddhist Sogdian, (MacKenzie 1970: 32, l. 552)  
 zk pwstk ywn'k 'cw n'm kyr't  
 DEF book this what name be.made.PRS.SUBJ.3SG  
 'What name should be given to this sūtra?'
- b. Christian Sogdian, Sims-Williams (1985: 129, 61R 26)  
 fr(y)t't pr() cw γrb't bwt  
 love PREP what known is  
 'By what can love be known?'
- Clause-initial wh
- c. Manichean Sogdian, (Sundermann 1985: 30, l. 170-171)  
 c'nkwy ZY myδ prw pt'yδy(h) [ZY]  
 how PRT so in shame and  
 pw βr'ty mwrtyš  
 without brother.OBL you.died  
 'How did you die in such shame and without the brother?'

Furthermore, in attested Sogdian and Middle Persian texts, wh-questions with non-preverbal wh-words were fairly rare (although still attested). I take it as evidence that the emergence of the preverbal focus was underway.

Taking into account that the emergence of a preverbal focus<sup>18</sup> and the relocation of negation into a preverbal position proceeded simultaneously, it is reasonable to assume that at this stage, negation still got adjoined to FocP (40).

- (40) XP ... XP [FocP Neg [FocP [V ... ]...]

<sup>18</sup> A reviewer inquires about how the preverbal placement of focus was achieved in Middle Iranian. It is quite plausible that it involved verb movement to Foc<sup>0</sup>, but the surviving material is too fragmentary to verify this assumption.

Such a system must have served as the grammaticalization source for NegP across Iranian languages, including an ancestor of Ossetic. Attested Middle Iranian languages developed the pattern ‘preverbal negation + negative polarity items’. This was so in Khotanese, Middle Persian, and Parthian, and to a large extent in Sogdian, (41). The latter did have true negative indefinites as well, albeit very rarely used, Sims-Williams (1986).

- (41) Sogdian
- a. **cnn** **’δ’** **s fryty** **’ty** **’pryty** **n’** **xcy**  
 from anything create.PRF.PRT and bless.PRF.PRT NEG be.PRS.3SG  
 “has not been created and blessed by anybody” (M264aR) Sundermann 1981
- b. **tym** **prm** **’’δk** **L’** **’’γt**  
 still until.now someone NEG arrive  
 ‘No one arrived until now.’ Mugh Letter A14: 26; Livshits (2008: 96)

Sogdian negative indefinites (42) are clearly an isolated development. Morphologically, they are not based on *wh*-stems, unlike the Old Iranian ‘no one’ or the system in Modern Ossetic, and, furthermore, they occur fairly rarely<sup>19</sup> in the surviving texts (Sims-Williams 1986).

- (42) Sogdian
- a. **ny-δk** **’’yst**  
 NEG-IDF comes  
 ‘No one comes’ Ancient Letters 2:8 (Sims-Williams 1986: 409)
- b. **ny-δ’cw** **ZK** **wr’kh** **ny-st**  
 NEG-IDF DEF success NEG-be.PRS.3SG  
 ‘there is no success’ (MacKenzie 1970: 22 v. 397)
- c. **rty** **k’w** **pwt’n’kδrm** **ny-δ’yc** **ZK** **wr’kh** **’krtw**  
 and to Buddha dharma NEG-IDF DEF success do.POT.3SG  
 ‘...can make no profit to the Buddha-dharma’ (MacKenzie 1976: 22 l.50)

This line of development disappeared with the extinction of Sogdian. Its quasi-descendant, Yaghnobi, only uses negative polarity items loaned from Tajiki, (Khromov 1972: 28, Andreev and Peshchereva 1957: 242).

Table 2 summarizes the developments across Middle Iranian languages.

Table 2. Evolution of Focus and negation system across across Middle Iranian languages

|                | Loss of Foc-fronting | Loss of Neg fronting | Emergence of Preverbal Focus | Emergence of Preverbal negation | Emergence of NPIs | Emergence of Neg-words |
|----------------|----------------------|----------------------|------------------------------|---------------------------------|-------------------|------------------------|
| Middle Persian | +                    | +                    | +                            | +                               | +                 | -                      |
| Parthian       | +                    | +                    | +                            | +                               | +                 | -                      |
| Sogdian        | +                    | +                    | +                            | +                               | +                 | + (disappear later)    |
| Khotanese      | +                    | Incomplete loss      | +                            | +                               | +                 | -                      |

We will now leave the rest of Iranian languages aside and concentrate on Ossetic-specific developments. First of all, the considerations above are pertinent to the discussion of Ossetic, because some synchronic evidence exists that negative markers used to be fronted in its ancestor as well. The evidence comes from the behavior of phasal adverbs =*bal* (in Digor)/ =*wal* (in Iron) ‘while’ and =*ma*

<sup>19</sup> About 15 occurrences in all published Sogdian texts according to the count of Sims-Williams (1986: 408, 421).

‘still/yet’<sup>20</sup>. Normally, they behave as second position clitics. For instance, in (43), they are placed after the first word in the respective clauses.

- (43) a. tso=**bal** talanʃi-mə  
go.IMP.2SG=while Talanchi-ALL  
‘In the meanwhile, go to Talanchi.’ Aghuzarti (2008: 86)
- b. ba-jzadəj=**ma**=nəmə kədin χussən  
PRV-remain.PST.3SG=still=ALL.1PL woodenbed  
‘We still owned a wooden bed.’ Aghuzarti (2008: 29)

When negation is present, however, these clitics can only attach to preverbal negative items, as the contrast between (44 a, c) and (44 b, d) shows. In the ungrammatical sentences in (44 b, d), =*bal* and =*ma* occupy the would-be normal second position in the clause, while in (44 a, c) they encliticize to the negative marker or a negative indefinite.

- (44) a. Soslan Mədin-i tuχχej [**ne-ke-bal** χətsʃə] fə-dʒdʒoruj  
S. M-GEN about NEG-who-more with PRV-talk.PRS.3SG  
‘Soslan does not talk about Madina to anybody anymore.’
- b. \*Soslan=**bal** [**neke** χətsʃə] Mədin-i tuχχej fə-dʒdʒoruj  
S.=while NEG-who with M-GEN about PRV-talk.PRS.3SG
- c. tukd **nə=ma** ba-ntsədəj  
war NEG=yet PRV-cease.PST.3SG  
‘The war has not yet ceased.’ Aghuzarti
- d. \*tukd=**ma** **nə** ba-ntsədəj  
war=yet NEG PRV-cease.PST.3SG  
*Idem* (intended)

A likely explanation of this phenomenon is that it is a survival from the time when negative markers were fronted.

Now, on the assumption that NegP was not grammaticalized by the period when ancestors of Ossetians settled in the North Caucasus, the emergence of the peculiar morphosyntax of Ossetic negation can be explained by contact influences from South Caucasian languages. For expository purposes, I will first describe the hypothesized stages of development in Ossetic in Sections 7.3-7.5, and then adduce South Caucasian evidence to support this scenario in Section 8.

### 7.3 Stage III: Obligatory movement of indefinites in the scope of negation

In Proto-Ossetic, while negation adjoined to the now preverbal FocP as in (40), indefinites in the scope of negation had to move into the specifiers of FocP. This movement could be still driven by a focus feature. It is only the negative marker that carried an [iNeg] feature.

- (45) [fronted material....Neg<sup>[iNeg]</sup>=<sub>[FocP]</sub> Id<sub>f1</sub> Id<sub>f2</sub> [V...] ... ]

Some synchronic evidence exists that bare *wh*-words served as wide-purpose indefinites in Proto-Ossetic, although now Ossetic has dedicated suffix- or prefix-marked *wh*-based indefinites<sup>21</sup>. At least one construction exists where bare *wh*-words function as non-interrogative indefinites with the meaning “some members of a certain group”, (46). This can be considered as a remnant of an earlier wider use of bare *wh*-words as indefinites.

<sup>20</sup> The homophony with *ma* NEG.IMP is accidental.

<sup>21</sup> See Haspelmath (1997: 281-282) for a list and description of functions of Iron Ossetic indefinites; the situation in Digor is very similar.

- (46) **ka**=ʃi            zar-gɐ            kɐnuj            **ka**=ba            ɐxɛrdigonaw    was-gɐ  
 who=ABL.3PL    sing-CNV            do.PRS.3SG            who=CTR            in.own.manner    twitter-CNV  
 ‘Some of them (birds) are singing, whereas some are twittering in their own manner.’  
 Xoxojti 2011: 135

#### 7.4 Stage IV: Split of preverbal FocP, emergence of NegP

At this stage, the focusing of negation and negative items was re-analyzed into movement into a separate syntactic position, with a phonologically null operator carrying a [iNeg] feature. This is similar to the emergence of wh-fronting from focalization of wh-phrases in early Indo-European languages, Kiparsky (1994). Indefinites in the scope of negation got endowed with a [uNeg] feature responsible for their movement, while Neg<sup>0</sup> got endowed with a matching strong feature. The negative marker, from a [iNeg]-adverb got re-analyzed into [uNeg] affix realized on the first of negative indefinites in the chain as schematically shown in (47).

- (47) a. [fronted material....Neg<sup>[uNeg]</sup>=<sub>FocP</sub> wh<sub>1</sub> wh<sub>2</sub> [VP] ... ]  
 b. Hypothetical Proto-Ossetic example  
       ɬɐw-i            [nɐ            ka            ʃi            iʁusta]  
       village-LOC    NEG            who        what        heard  
       [TopP            [FocP    [FocP                    ]...]  
       ‘No one heard anything in the village.’

At this stage, with the reanalysis of sentential negation into a dummy element in Spec NegP, a negative operator must have emerged high in the clause.

As for movement of wh-phrases in wh-questions, as was argued in Erschler (2012), it also underwent a change from focusing to grammaticalized wh-movement into a dedicated projection, the WP of (16) in Section 3.2. It is reasonable to assume that this development occurred simultaneously with the grammaticalization of the NegP and of negative movement.

#### 7.5 Stage V: Neg-feature receives a spell-out

At stage IV, the situation became asymmetric: although all negative indefinites carried [uNeg]-feature, it is only the first one in the chain that carried a negative prefix.

- (48) [fronted material....Neg<sup>[uNeg]</sup>-wh<sub>1</sub> wh<sub>2</sub><sup>[uNeg]</sup> [VP] ... ]  
       ɬɐw-i            [ne-ka            ʃi            iʁusta]  
       village-LOC    NEG-who        what        heard  
       [TopP            [NegP  
       ‘No one heard anything in the village.’

Then, an analogy-driven development occurred: the negative feature on negative indefinites received a spellout, which resulted in the negative prefix being copied to all negative indefinites present in the clause<sup>22</sup>.

- (49) [fronted material....Neg<sup>[uNeg]</sup>-wh<sub>1</sub> Neg<sup>[uNeg]</sup>-wh<sub>2</sub> [VP] ... ]  
       ɬɐw-i            [ne-ke            ne-ʃi            iʁusta]  
       village-LOC    NEG-who        NEG-what        heard  
       [TopP            [NegP  
       ‘No one heard anything in the village.’ (Digor Ossetic)

<sup>22</sup> A question remains of why the non-indicative negative prefix, *ma-*, is fully identical with the non-indicative sentential negation marker, while the indicative negative prefix, *ne-* (Digor)/*ni-* (Iron), and the indicative sentential negation marker *nɐ* exhibit different vowels. This phenomenon has no direct analog in Georgian or Svan. I leave this matter for further research.

The fact that the negative movement does not coincide with focusing *synchronously* in Modern Ossetic follows from the fact that while focused items may appear in the postverbal position (15d), negative indefinites may not do so, (19c), repeated here as (50a) and (50b) for the convenience of the reader. Focussed items, on the other hand, may appear postverbally in negated sentences (50c-d)<sup>23</sup>.

- Table 3 recapitulates the diachronic development scenario proposed in this section.

| Stage                                                                 | Position of sentential negation                             | Morphology of indefinites in the scope of negation                               | Position of indefinites in the scope of negation |
|-----------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------|
| Stage I; Old Indo-Iranian languages                                   | Negative Adverb; (can be) adjoined to sentence-initial FocP | Except for 'no one', intrinsically non-negative indefinites                      | Optional fronting to FocP                        |
| Stage II; Innovative Old Iranian and archaic Middle Iranian languages | Negative Adverb; (can be) adjoined to preverbal FocP        | Intrinsically non-negative indefinites                                           | Optional movement to FocP                        |
| Stage III; Proto-Ossetic                                              | Negative Adverb; obligatorily adjoined to preverbal FocP    | Intrinsically non-negative indefinites                                           | Obligatory movement to FocP                      |
| Stage IV; Proto-Ossetic                                               | Dummy element in Spec NegP                                  | Intrinsically non-negative indefinites and Neg-words with overt negative marking | Obligatory movement to NegP                      |
| Stage V, Modern Ossetic                                               | Dummy element in Spec NegP                                  | Neg-words with overt negative marking                                            | Obligatory movement to NegP                      |

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## 8. South Caucasian evidence

While Stages III-V described above are fully hypothetical in the case of Ossetic, more direct evidence for a development of this type can be marshalled from South Caucasian languages, specifically, Georgian and Svan.

Written records of Georgian allow us to track this development. Old Georgian used two strategies of dealing with indefinites in the scope of negation. The negative marker could either precede the cluster of indefinites, as shown in (51a), which corresponds to Stage III or IV above, or the negative marker was copied on each of the indefinites, (51b). The prefixes *ara* and *nu* are homonymous with the sentential negation markers.

- (51) Old Georgian
- a. **ara**-raj vis utxres  
 NEG-what who.DAT they.told  
 ‘They told nothing to anybody’ (Lk 9: 36)
- b. **nu**-sada **nu**-ra-s borot’-sa ikm-t  
 N.IMP-where N.IMP-what-DAT wicked-DAT do.IMP-2PL  
 ‘Never do anything (lit. nothing) (so) wicked.’ (Judges 19:23)

On the other hand, Modern Georgian requires a negative prefix on all indefinites, as is illustrated by the contrast between the grammatical sentences in (52a) and the ungrammatical ones in (52b), where the negative marker is missing from some of the indefinites. That is to say, Georgian reached Stage V from Section 6.5 above, as far as the morphology of indefinites in the scope of negation is concerned.

- (52) Modern Georgian
- a. porum-ze **ara**-vin **ara**-sodes **ara**-pers (ar) afavebs  
 forum-on NEG-who NEG-ever NEG-color NEG do.wrong  
 ‘On the forum, no one ever commits anything wrong’<sup>24</sup>
- b. \*porum-ze **ara**-vin sodes pers (ar) afavebs  
 forum-on NEG-who ever color NEG commit.wrong  
*Idem* (intended)

One clear difference exists between Modern Georgian and Ossetic, the possibility for a sentential negator to co-occur with negative indefinites. This, however, is a later innovation in Georgian, as was shown by Topuria (1923/2002).

A system analogous to Old Georgian still obtains in Svan – the preverbal chain of indefinites in the scope of negation may either consist of Neg-words (53a), or the negative marker is put at the beginning of the chain (53b-c).

- (53) Svan, Erschler (2022)
- a. XP NI ... NI (\*Neg) V (XP)  
 gela-d **de:-mʃik** **de:-me** **d-æ:r**  
 G-ERG NEG-ever NEG-where NEG-who  
 (\***de:m/\*ma:m**) anundrevi:ne  
 NEG offended  
 ‘Gela has never offended anyone anywhere.’
- b. XP NI NPI ... NPI V (XP)  
 gela-d **de:-mʃik** **ime** **jæ:r** anundrevi:ne  
 G-ERG NEG-ever where who offended  
 ‘Gela has never offended anyone anywhere.’

<sup>24</sup> <https://forum.ge/?showtopic=33762819>, accessed 04/10/2023.

- c.      lædi    **ma:mgweʃ**    jæ:r-d      anmeqre  
          today   nothing                who-ERG      understood  
          ‘Today no one understood anything.’

Accordingly, Svan allows us to take a glimpse into the system as it apparently functioned in Old Georgian. On the other hand, the documented development from Old to Modern Georgian allows us to see how copying of negative morphology on each indefinites becomes obligatory, as in the transition from Stage IV to Stage V of the scenario proposed here in Ossetic. I assume that the same development occurred in Ossetic.

## 9. An alternative scenario

A possible alternative scenario is that the negative feature got spelled out on indefinites prior to their relocation to the preverbal position, somewhat analogously to what was happening in Sogdian, as described in (42) Section 7.2. A conjectural course of events under this scenario is schematically shown in (54).

- (54) a.      Stage A  
               (fronted material) NEG-wh<sub>1</sub> XP NEG-wh<sub>2</sub> YP NEG V  
       b.      Stage B  
               (fronted material) XP NEG-wh<sub>1</sub> NEG-wh<sub>2</sub> NEG V  
       c.      Stage C  
               (fronted material) XP NEG-wh<sub>1</sub> NEG-wh<sub>2</sub> (NEG) V  
       d.      Stage D  
               (fronted material) XP NEG-wh<sub>1</sub> NEG-wh<sub>2</sub> (\*NEG) V

If that was the course of events, we would need a separate explanation for the emergence of obligatory negative spread in Ossetic. A probable scenario would be weakening and disappearance of Neg as a stage in Jespersen cycle, however, no documented examples of such a development seem to exist in any language.

The reverse development, replacement of negative spread by strict negative concord is well attested: negative spread was possible in Early Slavic, but then Strict Negative Concord became obligatory (Willis 2013: 370-373 and references there). As we have already seen, the same occurred in Georgian, Topuria (1923/2002). More recently, the same development happened under the influence of Modern Georgian in the dialects of Iron Ossetic spoken in South Ossetia, (Bekoev 1985: 236, Dzigcojty 2008: 71, Erschler 2018: 878). Therefore, an alternative scenario of this type appears implausible.

## 10. Conclusion

The Modern Ossetic languages exhibit a typologically uncommon pattern of negative concord – a combination of obligatory negative spread and obligatory movement of Neg-phrases into an immediately preverbal position. In this paper, I have argued that this system developed historically from a fairly common phenomenon, the preverbal placement of focus. Under the proposed scenario, the negative movement is diachronically related to focusing, whereas the negative spread is an outcome of the fact that the negation markers got reanalyzed into the negative prefixes on wh-items.

The proposed scenario is another example of how a word order pattern initially motivated by requirements of information structure grammaticalizes into an independent syntactic process. This idea was invoked in the literature to explain the emergence of a wide range of grammatical phenomena, (Sankoff and Brown 1976, Givón 1979, Shibatani 1991, Lehmann 2008).

Methodologically, this paper is an attempt at syntactic reconstruction in the absence of direct diachronic evidence. To overcome the lack of Ossetic historical data, I have used two types of proxies. To reconstruct earlier stages of development of the negation system, which are arguably largely identical across the Iranian, I drew upon the attested Old and Middle Iranian languages. For later stages of development, which are Ossetic-specific, I turned to unrelated, but geographically close

South Caucasian languages, which have demonstrably exercised considerable influence on Ossetic grammar and which are typologically similar to Ossetic in the relevant aspects. Hopefully, this method can be applied to investigate other cases of language change under language contact.

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